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(73) Proprietors:

- **Merck Sharp & Dohme LLC**
Rahway, New Jersey 07065-0907 (US)
- **MSD R&D (China) Co., Ltd.**
Beijing 100020 (CN)
- **THE WALTER AND ELIZA HALL INSTITUTE OF MEDICAL RESEARCH**
Parkville, VIC 3052 (AU)

(72) Inventors:

- **MCCAULEY, John A.**
West Point, Pennsylvania 19486 (US)
- **DE LERA RUIZ, Manuel**
West Point, Pennsylvania 19486 (US)

- **GUO, Zhuyan**
West Point, Pennsylvania 19486 (US)
- **HU, Bin**
Shanghai 200131 (CN)
- **KELLY III, Michael J.**
West Point, Pennsylvania 19486 (US)
- **LEI, Zhiyu**
Shanghai 200131 (CN)
- **OLSEN, David B.**
West Point, Pennsylvania 19486 (US)
- **SLEEBES, Brad**
Parkville, Victoria 3052 (AU)
- **ZHAN, Dongmei**
Shanghai 200131 (CN)
- **ZHAO, Lianyun**
Shanghai 200233 (CN)

(74) Representative: **Merck Sharp & Dohme LLC**

120 Moorgate
London EC2M 6UR (GB)

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Description

FIELD OF THE INVENTION

[0001] The present invention relates to compounds of Formula (I), or pharmaceutically acceptable salts thereof, useful for the treatment of *Plasmodium* infections. More specifically, the present invention relates to compounds of Formula (I), or pharmaceutically acceptable salts thereof, useful for the treatment of *Plasmodium* infections, more particularly for the treatment of malaria.

BACKGROUND OF THE INVENTION

[0002] Malaria is a major disease in humans with several hundred million infections and over 450,000 deaths each year. The most lethal form of malaria is caused by *Plasmodium falciparum*. This protozoan parasite is responsible for almost all malarial deaths with most occurring in Africa. *P. falciparum* has a complex life cycle starting in the Anopheles mosquito vector when sporozoite forms are injected into the human host during a blood feed. These sporozoites migrate to the liver and invade hepatocytes in which they develop to form thousands of liver merozoites that egress into the blood where they invade erythrocytes to commence the asexual cycle of the parasite responsible for the symptoms of malaria. The parasite develops within the protected niche of the red cell to form 16-32 merozoites that, once mature, egress from the host cell to invade new red blood cells. Some of these parasites differentiate to form gametocytes, the sexual form of the parasite. These can be taken up by the mosquito where male and female gametes form, fuse and differentiate into oocysts on the mosquito midgut extracellular matrix. Sporozoites form within the oocyst and upon egress migrate to the salivary gland for delivery to the next host during blood feeding for perpetuation and survival of the parasite.

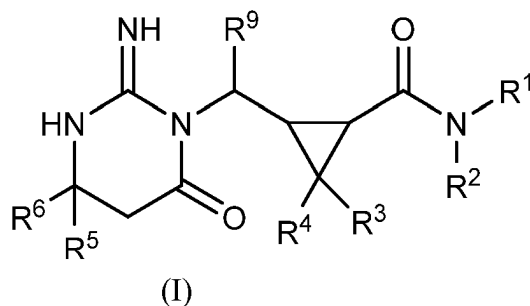
[0003] Other forms of malaria include a relapsing form of malaria caused by *P. vivax* which is responsible for significant morbidity, can cause virulent forms of this disease with some deaths and is mainly a problem outside Africa. *P. knowlesi* is found in South East Asia and is a zoonotic parasite that normally infects long-tailed macaques but has been shown to infect humans in Malaysian Borneo.

[0004] Artemisinin combined with partner drugs have become a mainstay in the treatment and control of malaria. Currently, the WHO recommend artemisinin-based combinations (ACTs) for treatment of uncomplicated malaria caused by *P. falciparum*. These are very effective; however, ACT resistance is now leading to treatment failures in the Greater Mekong Subregion (GMS) and the prospect of spread outside this region is extremely concerning. However, due to the increasing threat of artemisinin-based combination therapy (ACT) drug resistance, the development of new antimalarials with novel targets that inhibit multiple steps in the parasite life cycle is an urgent priority for the malaria control field. Such novel antimalarials, as monotherapies or ACT partner drugs, could make strides towards malaria elimination as there is a reduced likelihood of parasites with preexisting resistance mutations being present in the parasite population.

[0005] Currently, aspartic acid proteases are prime targets for drug development: the HIV aspartic acid protease has been successfully targeted with a drug in clinical use; inhibitors that target human renin, BACE1 and gamma-secretase have been or are in clinical development. In the antimalarial drug space, *P. falciparum* aspartic acid proteases plasmepsin X and IX (PMX and PMIX) have been identified as potential targets since inhibitors block parasite egress and invasion of the host cell and prevent maturation of some rhoptry and micronemal proteins required for this process (Pino P, Caldelari R, Mukherjee B, Vahokoski J, Klages N, Maco B, et al. A multistage antimalarial targets the plasmepsins IX and X essential for invasion and egress. Science. 2017;358(6362):522-8.)

SUMMARY OF THE INVENTION

[0006] The present invention is directed to compounds of Formula (I):



wherein R¹, R², R³, R⁴, R⁵, R⁶ and R⁹ are described below.

[0007] Also described herein are methods of treatment of *Plasmodium* infections comprising administering to a subject in need thereof a compound of Formula (I), or a pharmaceutically acceptable salt thereof. Also described herein are methods of treatment of *Plasmodium* infections comprising administering to a subject in need thereof a compound of Formula (I), or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable carrier. A "method of treatment" refers to a compound of formula (I) for use in said treatment.

[0008] Also described herein are methods of treating of malaria comprising administering to a subject in need thereof a compound of Formula (I), or a pharmaceutically acceptable salt thereof.

[0009] The present invention further provides the use of compositions, including pharmaceutical compositions, comprising one or more compounds of the invention (e.g., one compound of the invention), or a tautomer thereof, or a pharmaceutically acceptable salt or solvate of said compound(s) and/or said tautomer(s), optionally together with one or more additional therapeutic agents, optionally in an acceptable (e.g., pharmaceutically acceptable) carrier or diluent, for the treatment of malaria.

[0010] Moreover, the present invention provides methods for the use of pharmaceutical compositions comprising one or more of said compounds in the free form or in pharmaceutically acceptable salt form, together with one or more customary pharmaceutical excipient(s), for the treatment of *Plasmodium* infections, the treatment of malaria, the inhibition of plasmepsin X, or, possibly in some embodiments, the dual inhibition of plasmepsin X and plasmepsin IX. Methods for the use of combinations of the compounds or salts of the invention together with one or more additional pharmaceutically active agents are also provided.

[0011] The present invention may further provide methods for the inhibition of plasmepsin X, or dual inhibition of plasmepsin X and plasmepsin IX activity and for treatment, prevention, amelioration and/or delaying onset of diseases or disorders in which the inhibition of plasmepsin X and/or plasmepsin IX has or may have a therapeutic effect, e.g. malaria.

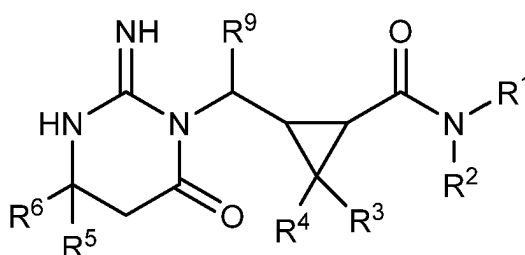
[0012] The present invention further provides methods for the inhibition of *P. falciparum* aspartic acid proteases. The present invention may further provide methods for blocking *P. falciparum* growth by inhibiting Plasmepsin X. The present invention may further provide methods for blocking *P. falciparum* growth by inhibiting both Plasmepsin X and Plasmepsin IX. In certain embodiments, the present invention may provide a method for blocking *P. falciparum* growth in an animal, preferably a human, comprising administering to the animal an inhibitor of Plasmepsin IX and Plasmepsin X or a pharmaceutically acceptable salt thereof.

[0013] The present invention may further provide methods for the treatment of malaria by inhibiting plasmepsin X. The present invention may further provide methods for the treatment of malaria by inhibiting both Plasmepsin X and Plasmepsin IX. In certain embodiments, the present invention may provide a method for treating malaria in an animal, preferably a human, comprising administering to the animal an inhibitor of Plasmepsin IX and Plasmepsin X or a pharmaceutically acceptable salt thereof.

[0014] These and other embodiments of the invention, which are described in detail below or will become clear to those of ordinary skill in the art, are included within the scope of the invention.

DETAILED DESCRIPTION OF THE INVENTION

[0015] Described herein are compounds having the structural Formula (I):



(I)

or a pharmaceutically acceptable salt thereof, wherein:

R¹ is a heterocycloalkyl, C₃-C₁₂cycloalkyl, aryl, C₁-C₆alkylaryl, wherein the heterocycloalkyl, C₃-C₁₂cycloalkyl, aryl or C₁-C₆alkylaryl is unsubstituted or substituted with 1 to 5 substituents independently selected from the group consisting of halogen, CN, OH, alkoxy, haloalkoxy, C₁-C₆alkylOC₁-C₆alkyl, C₁-C₆alkylCOOH, COOH, oxo, COOC₁-C₆alkyl, C₃-C₆cycloalkyl, spiroC₃-C₆cycloalkyl, C₁-C₆alkyl, haloC₁-C₆alkyl, C₁-C₆alkylOH, CON(R⁷)(R⁸), N(R⁷)(R⁸) and C₁-C₆alkylN(R⁷)(R⁸);

R² is hydrogen, C₁-C₆alkylCOOH, COOH, C₃-C₆cycloalkyl, C₁-C₆alkyl, haloC₁-C₆alkyl or C₁-C₆alkylOH;

R³ is hydrogen, halogen, CN, OH, C₃-C₆cycloalkyl or C₁-C₆alkyl;

R⁴ is hydrogen, halogen, CN, OH, C₃-C₆cycloalkyl or C₁-C₆alkyl;

R⁵ is hydrogen, halogen, CN, OH, alkoxy, C₁-C₆alkylOC₁-C₆alkyl, C₁-C₆alkylCOOH, COOH, C₃-C₆cycloalkyl, C₁-C₆alkyl, haloC₁-C₆alkyl, C₁-C₆alkylOH, CON(R⁷)(R⁸), N(R⁷)(R⁸) or C₁-C₆alkylN(R⁷)(R⁸) or when taken with R⁶ forms a C₃-C₆cycloalkyl or C₃-C₆heterocycloalkyl;

R⁶ is hydrogen, halogen, CN, OH, alkoxy, C₁-C₆alkylOC₁-C₆alkyl, C₁-C₆alkylCOOH, COOH, C₃-C₆cycloalkyl, C₁-C₆alkyl, haloC₁-C₆alkyl, C₁-C₆alkylOH, CON(R⁷)(R⁸), N(R⁷)(R⁸) or C₁-C₆alkylN(R⁷)(R⁸) or when taken with R⁵ forms a C₃-C₆cycloalkyl or C₃-C₆heterocycloalkyl;

R⁷ is hydrogen, C₁-C₆alkylCOOH, COOH, C₃-C₆cycloalkyl, C₁-C₆alkyl, haloC₁-C₆alkyl or C₁-C₆alkylOH;

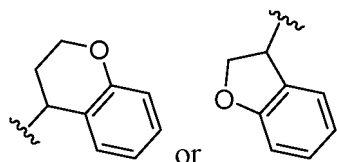
R⁸ is hydrogen, C₁-C₆alkylCOOH, COOH, C₃-C₆cycloalkyl, C₁-C₆alkyl, haloC₁-C₆alkyl or C₁-C₆alkylOH; and

R⁹ is hydrogen, halogen, CN, alkoxy, C₁-C₆alkyl, heterocycloalkyl, heteroaryl, C₃-C₁₂cycloalkyl, aryl, COOH, oxo, COOC₁-C₆alkyl, haloC₁-C₆alkyl, OH, CON(R⁷)(R⁸) and N(R⁷)(R⁸), wherein the C₁-C₆alkyl can be unsubstituted or substituted with one, two or three substituents selected from the group consisting of halogen, CN, OH, alkoxy, heterocycloalkyl, heteroaryl, C₃-C₁₂cycloalkyl, aryl, COOH, oxo, COOC₁-C₆alkyl, haloC₁-C₆alkyl, OH, CON(R⁷)(R⁸) and N(R⁷)(R⁸), and wherein the heterocycloalkyl, heteroaryl, C₃-C₁₂cycloalkyl and aryl can be unsubstituted or substituted with one, two or three substituents selected from the group consisting of halogen, CN, OH, alkoxy, C₁-C₆alkyl, COOH, oxo, COOC₁-C₆alkyl, haloC₁-C₆alkyl, OH, CON(R⁷)(R⁸) and N(R⁷)(R⁸).

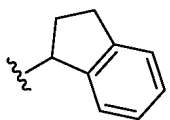
[0016] In the embodiments described herein, R¹ is a heterocycloalkyl, C₃-C₁₂cycloalkyl, aryl, C₁-C₆alkylaryl, wherein the heterocycloalkyl, C₃-C₁₂cycloalkyl, aryl or C₁-C₆alkylaryl is unsubstituted or substituted with 1 to 5 substituents independently selected from the group consisting of halogen, CN, OH, alkoxy, haloalkoxy, C₁-C₆alkylOC₁-C₆alkyl, C₁-C₆alkylCOOH, COOH, oxo, COOC₁-C₆alkyl, C₃-C₆cycloalkyl, spiroC₃-C₆cycloalkyl, C₁-C₆alkyl, haloC₁-C₆alkyl, C₁-C₆alkylOH, CON(R⁷)(R⁸), N(R⁷)(R⁸) and C₁-C₆alkylN(R⁷)(R⁸).

[0017] In certain embodiments, R¹ is a bicyclic ring. In certain embodiments, R¹ is a bicyclic heterocycloalkyl, bicyclic C₃-C₁₂cycloalkyl or bicyclic aryl ring. In certain embodiments, R¹ is a fused bicyclic ring having an A ring fused with a B ring, wherein the A ring is a 5, 6 or 7-membered saturated ring and the B ring is a phenyl or a 5 or 6-membered heteroaryl ring. In certain embodiments, the A ring includes at least one O, N, or S. In certain embodiments, the A ring includes at least one O.

[0018] In certain embodiments, R¹ is a heterocycloalkyl. Non-limiting examples of monocyclic heterocycloalkyl groups include piperidyl, oxetanyl, pyrrolyl, piperazinyl, morpholinyl, thiomorpholinyl, thiazolidinyl, 1,4-dioxanyl, tetrahydrofuran, tetrahydrothiophenyl, *beta* lactam, *gamma* lactam, *delta* lactam, *beta* lactone, *gamma* lactone, *delta* lactone, and pyrrolidinone, and oxides thereof. Non-limiting examples of bicyclic heterocycloalkyl groups include, but are not limited to,

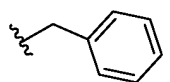


[0019] In certain embodiments, R¹ is a C₃-C₁₂cycloalkyl. In certain embodiments, the cycloalkyl is a monocyclic cycloalkyl. Suitable examples of cycloalkyls include, but are not limited to, cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl. In certain embodiments, the cycloalkyl is a bicyclic cycloalkyl. Suitable examples of cycloalkyls include, but are not limited to:



[0020] In certain embodiments, R¹ is an aryl ring. Suitable examples of aryls include, but are not limited to, monocyclic aryl groups such as, phenyl and bicyclic aryl groups such as naphthyl.

[0021] In certain embodiments, R¹ is a C₁-C₆alkylaryl ring. Suitable examples of C₁-C₆alkylaryls include, but are not limited to:



[0022] In certain embodiments, R^1 is a chomane or indane.

[0023] In certain embodiments, R^1 is unsubstituted. In other embodiments, R^1 is substituted with 1 to 5 substituents independently selected from the group consisting of halogen, CN, OH, alkoxy, haloalkoxy, C_1 - C_6 alkylOC $_1$ - C_6 alkyl, C_1 - C_6 alkylCOOH, COOH, oxo, COOC $_1$ - C_6 alkyl, C_3 - C_6 cycloalkyl, spiro C_3 - C_6 cycloalkyl, C_1 - C_6 alkyl, halo C_1 - C_6 alkyl, C_1 - C_6 alkylOH, CON(R^7)(R^8), N(R^7)(R^8) and C_1 - C_6 alkylN(R^7)(R^8). In certain embodiments, R^1 is substituted with 1 substituent. In certain embodiments, R^1 is substituted with 2 substituents. In certain embodiments, R^1 is substituted with 3 substituents. In certain embodiments, R^1 is substituted with 4 substituents. In certain embodiments, R^1 is substituted with 5 substituents.

[0024] In certain embodiments, R^1 is substituted with halogen. Examples of suitable halogens include, but are not limited to, chlorine, bromine, fluorine and iodine. In certain embodiments, R^1 is substituted with CN. In certain embodiments, R^1 is substituted with OH. In certain embodiments, R^1 is substituted with an oxo group.

[0025] In certain embodiments, R^1 is substituted with alkoxy. Suitable alkoxys include, but are not limited to, methoxy, ethoxy, n-propoxy, isopropoxy and n-butoxy. In certain embodiments, R^1 is substituted with C_1 - C_6 alkylOC $_1$ - C_6 alkyl. In certain embodiments, R^1 is substituted with haloalkoxy. Suitable haloalkoxys include, but are not limited to, trifluoromethoxy, fluoromethoxy and difluoromethoxy. In certain embodiments, R^1 is substituted with C_1 - C_6 alkylOC $_1$ - C_6 alkyl. Suitable C_1 - C_6 alkylOC $_1$ - C_6 alkyls include, but are not limited to, CH_2OCH_3 . In certain embodiments, R^1 is substituted with C_1 - C_6 alkylCOOH. In certain embodiments, R^1 is substituted with COOH. In certain embodiments, R^1 is substituted with C_1 - C_6 alkylCOOC $_1$ - C_6 alkyl.

[0026] In certain embodiments, R^1 is substituted with C_3 - C_6 cycloalkyl. Suitable examples of cycloalkyls include, but are not limited to, cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl. In certain embodiments, R^1 is substituted with cyclopropyl:



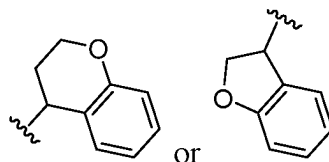
[0027] In certain embodiments, R^1 is substituted with spiro C_3 - C_6 cycloalkyl. Suitable examples of spirocycloalkyls include, but are not limited to, spirocyclopropyl, spirocyclobutyl, spirocyclopentyl and spirocyclohexyl. In certain embodiments, R^1 is substituted with spirocyclobutyl:



[0028] In certain embodiments, R^1 is substituted with C_1 - C_6 alkyl. Examples of C_1 - C_6 alkyl groups can include but are not limited to, methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, isopentyl, neopentyl, tert-pentyl, 1-methylbutyl, 2-methylbutyl, 1,2-dimethylpropyl, 1-ethylpropyl, n-hexyl, isohexyl, 1-methylpentyl, 2-methylpentyl, 3-methylpentyl, 1,1-dimethylbutyl, 1,2-dimethylbutyl, 2,2-dimethylbutyl, 1-ethylbutyl, 1,1,2-trimethylpropyl, 1,2,2-trimethylpropyl, 1-ethyl-2-methylpropyl and 1-ethyl-1-methylpropyl. In certain embodiments, R^1 is substituted with halo C_1 - C_6 alkyl. Suitable examples of haloalkyls include, but are not limited to, fluoromethyl, difluoromethyl, trifluoromethyl, 2-fluoroethyl, 1,2-difluoroethyl and 2,2-difluoroethyl. In certain embodiments, R^1 is substituted with C_1 - C_6 alkylOH. Examples of suitable alcohols, include, but are not limited to, methanol, ethanol, propanol, butanol and iso-butanol. In certain embodiments, R^1 is substituted with CON(R^7)(R^8). In certain embodiments, R^1 is substituted with N(R^7)(R^8). In certain embodiments, R^1 is substituted with C_1 - C_6 alkylN(R^7)(R^8), wherein R^7 and R^8 will be described in detail below.

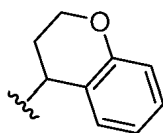
[0029] In certain embodiments, R^1 is substituted with 1 to 4 substituents selected independently from the group consisting of bromine, fluorine, chlorine, methyl, ethyl, t-butyl, methoxy, OH, CN oxo, CH_2OCH_3 , cyclopropyl, spirocyclobutyl, trifluoromethoxy and trifluoromethyl.

[0030] In certain embodiments, R^1 is



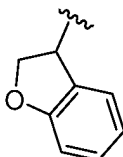
and is substituted with 1 to 4 substituents selected independently from the group consisting of bromine, fluorine, chlorine, methyl, ethyl, t-butyl, methoxy, OH, CN oxo, CH_2OCH_3 , cyclopropyl, spirocyclobutyl, trifluoromethoxy and trifluoromethyl.

[0031] In certain embodiments, R^1 is



and is substituted with 1 to 4 substituents selected independently from the group consisting of bromine, fluorine, chlorine, methyl, ethyl, t-butyl, methoxy, OH, CN oxo, CH_2OCH_3 , cyclopropyl, spirocyclobutyl, trifluoromethoxy and trifluoromethyl.

[0032] In certain embodiments, R^1 is



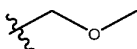
and is substituted with 1 to 4 substituents selected independently from the group consisting of bromine, fluorine, chlorine, methyl, ethyl, t-butyl, methoxy, OH, CN oxo, CH_2OCH_3 , cyclopropyl, spirocyclobutyl, trifluoromethoxy and trifluoromethyl. In certain embodiments described herein, R^2 is hydrogen, $\text{C}_1\text{-C}_6\text{alkylCOOH}$, COOH , $\text{C}_3\text{-C}_6\text{cycloalkyl}$, $\text{C}_1\text{-C}_6\text{alkyl}$, halo $\text{C}_1\text{-C}_6\text{alkyl}$ or $\text{C}_1\text{-C}_6\text{alkylOH}$. In certain embodiments, R^2 is hydrogen. In certain embodiments, R^2 is $\text{C}_1\text{-C}_6\text{alkylCOOH}$. In certain embodiments, R^2 is COOH . In certain embodiments, R^2 is $\text{C}_3\text{-C}_6\text{cycloalkyl}$. Suitable examples of cycloalkyls include, but are not limited to, cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl. In certain embodiments, R^2 is $\text{C}_1\text{-C}_6\text{alkyl}$. Examples of $\text{C}_1\text{-C}_6\text{alkyl}$ groups can include but are not limited to, methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, isopentyl, neopentyl, tert-pentyl, 1-methylbutyl, 2-methylbutyl, 1,2-dimethylpropyl, 1-ethylpropyl, n-hexyl, isohexyl, 1-methylpentyl, 2-methylpentyl, 3-methylpentyl, 1,1-dimethylbutyl, 1,2-dimethylbutyl, 2,2-dimethylbutyl, 1-ethylbutyl, 1,1,2-trimethylpropyl, 1,2,2-trimethylpropyl, 1-ethyl-2-methylpropyl and 1-ethyl-1-methylpropyl.

[0033] In certain embodiments, R^2 is halo $\text{C}_1\text{-C}_6\text{alkyl}$. Suitable examples of haloalkyls include, but are not limited to, fluoromethyl, difluoromethyl, trifluoromethyl, 2-fluoroethyl, 1,2-difluoroethyl and 2,2-difluoroethyl. In certain embodiments, R^2 is $\text{C}_1\text{-C}_6\text{alkylOH}$. Examples of suitable alcohols, include, but are not limited to, methanol, ethanol, propanol, butanol and iso-butanol.

[0034] With regard to the compounds described herein, R^3 is hydrogen, halogen, CN, OH, $\text{C}_3\text{-C}_6\text{cycloalkyl}$ or $\text{C}_1\text{-C}_6\text{alkyl}$. In certain embodiments, R^3 is hydrogen. In certain embodiments, R^3 is halogen. Suitable halogens include, but are not limited to, fluorine, chlorine, bromine, or iodine. In certain embodiments, R^3 is CN. In certain embodiments, R^3 is OH.

[0035] In certain embodiments, R^3 is $\text{C}_3\text{-C}_6\text{cycloalkyl}$. Suitable examples of cycloalkyls include, but are not limited to, cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl. In certain embodiments, R^3 is $\text{C}_1\text{-C}_6\text{alkyl}$. Examples of $\text{C}_1\text{-C}_6\text{alkyl}$ groups can include but are not limited to, methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, isopentyl, neopentyl, tert-pentyl, 1-methylbutyl, 2-methylbutyl, 1,2-dimethylpropyl, 1-ethylpropyl, n-hexyl, isohexyl, 1-methylpentyl, 2-methylpentyl, 3-methylpentyl, 1,1-dimethylbutyl, 1,2-dimethylbutyl, 2,2-dimethylbutyl, 1-ethylbutyl, 1,1,2-trimethylpropyl, 1,2,2-trimethylpropyl, 1-ethyl-2-methylpropyl and 1-ethyl-1-methylpropyl.

[0036] In certain embodiments, R^3 is hydrogen, fluorine, methyl, ethyl or



In certain embodiments, R^3 is taken with R^4 to form oxetanyl.

[0037] With regard to the compounds described herein, R^4 is hydrogen, halogen, CN, OH, $\text{C}_3\text{-C}_6\text{cycloalkyl}$ or $\text{C}_1\text{-C}_6\text{alkyl}$. In certain embodiments, R^4 is hydrogen. In certain embodiments, R^4 is halogen. Suitable halogens include, but are not limited to, fluorine, chlorine, bromine, or iodine. In certain embodiments, R^4 is CN. In certain embodiments, R^4 is OH.

[0038] In certain embodiments, R^4 is $\text{C}_3\text{-C}_6\text{cycloalkyl}$. Suitable examples of cycloalkyls include, but are not limited to, cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl. In certain embodiments, R^4 is $\text{C}_1\text{-C}_6\text{alkyl}$. Examples of $\text{C}_1\text{-C}_6\text{alkyl}$ groups can include but are not limited to, methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, isopentyl, neopentyl, tert-pentyl, 1-methylbutyl, 2-methylbutyl, 1,2-dimethylpropyl, 1-ethylpropyl, n-hexyl, isohexyl, 1-methylpentyl, 2-methylpentyl, 3-methylpentyl, 1,1-dimethylbutyl, 1,2-dimethylbutyl, 2,2-dimethylbutyl, 1-ethylbutyl, 1,1,2-trimethylpropyl, 1,2,2-trimethylpropyl, 1-ethyl-2-methylpropyl and 1-ethyl-1-methylpropyl.

[0039] In certain embodiments, R^4 is hydrogen, methyl, or fluorine. In certain embodiments, R^3 and R^4 are both

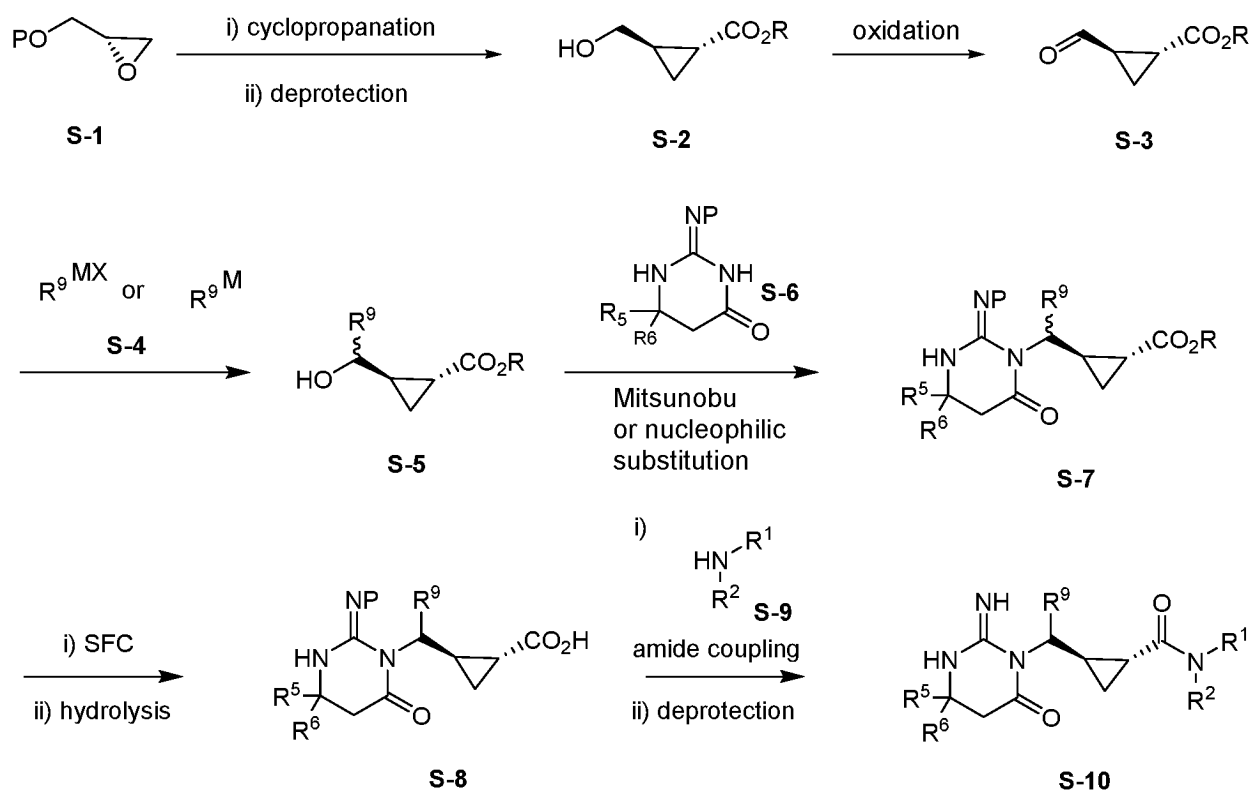
CD₃OD = heavy methanol

1 Standard atmosphere [atm] = 101325 pascal [Pa] = 14.6959488 psi

[0136] The meanings of the abbreviations in the nuclear magnetic resonance spectra are shown below: s = singlet, d = doublet, dd = double doublet, dt = double triplet, ddd = double double doublet, sept = septet, t = triplet, m = multiplet, br = broad, brs = broad singlet, q = quartet, J = coupling constant and Hz = hertz.

[0137] Several methods for preparing the compounds of this disclosure are described in the following Schemes and Examples. Starting materials and intermediates were purchased commercially from common catalog sources or were made using known procedures, or as otherwise illustrated. Some frequently applied routes to the compounds of Formula I are described in the Schemes that follow. In some cases, the order of carrying out the reaction steps in the schemes may be varied to facilitate the reaction or to avoid unwanted reaction products. An asterisk (*) may be used in a chemical structure drawing that indicates the location of a chiral center.

SCHEME 1



[0138] Intermediate compounds of Formula **S-2** are prepared from chiral epoxides **S-1**, in which P is an alcohol protecting group, after cyclopropanation conditions such as (EtO)₂POCH₂CO₂R/NaH followed by alcohol deprotection. Alcohol oxidation in **S-2** can be performed using oxidants such as Dess-Martin periodinane to yield aldehydes **S-3**. Treatment of **S-3** with an organometallic reagent **S-4**, in which M is a metal and X is a halogen, such as organozinc (M = Zn) and organomagnesium (M = Mg) species gives secondary alcohol intermediates **S-5**. Iminopyrimidones or iminohydantoins **S-6** are introduced under Mitsunobu conditions to give intermediates **S-7**. Alternatively, alcohol in intermediates **S-5** could be transformed into a leaving group such as a mesylate, tosylate, triflate or halogen which can be displaced with intermediates **S-6** to give intermediates **S-7**. At this step diastereomeric mixtures can be separated by SFC and the resulting chiral esters subjected to acid or base catalyzed hydrolysis or hydrogenation to give chiral acid intermediates **S-8**. Finally, **S-8** intermediates are coupled with amines **S-9** to provide final products of Formula **S-10** after iminopyrimidone deprotection using acids as TFA or Zn.

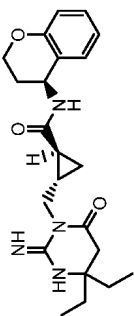
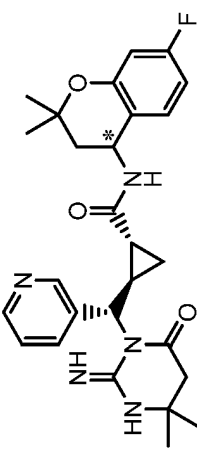
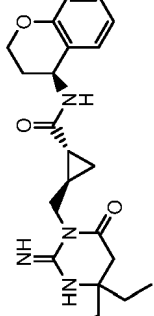
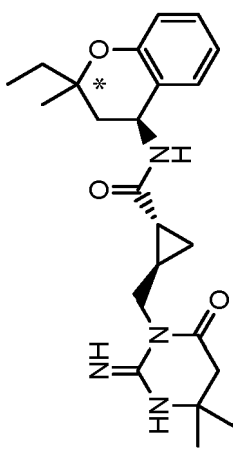
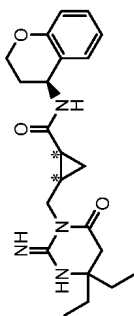
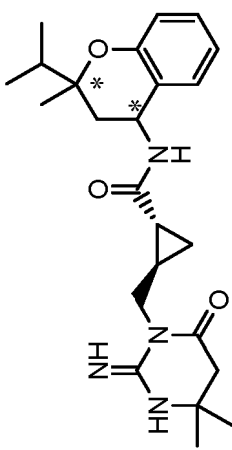
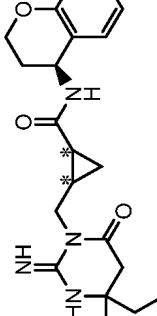
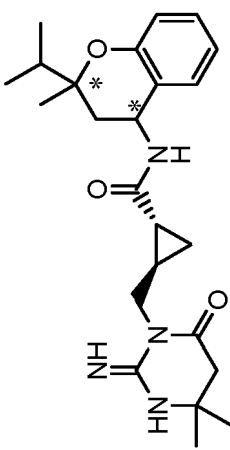
[0139] Reactions sensitive to moisture or air were performed inside a glove-box or under nitrogen or argon using anhydrous solvents and reagents. The progress of reactions was determined by either analytical thin layer chromatography (TLC) usually performed with E. Merck pre-coated TLC plates, silica gel 60F-254, layer thickness 0.25 mm or liquid chromatography-mass spectrometry (LC/MS).

[0140] Typically, the analytical LC-MS system used consisted of a Waters ZQ™ platform with electrospray ionization in

Table 1

Ex.	Structure	LC/MS (M+1)+	Ex.	Structure	LC/MS (M+1) +
1A		485.3	75C		471.3
1B		485.4	75D		471.3
2		504.3	76A		475.3
3A		371.1	76B		475.3
3B		371.3	77A		494.3

(continued)

Ex.	Structure	LC/MS (M+1)+	Ex.	Structure	LC/MS (M+1) +
4A		399.2	77B		494.3
4B		399.2	78		413.3
4C		399.3	79A		427.3
4D		399.2	79B		427.3